

论文笔记: Multigranularity Information Fused Contrastive Learning With Multiview Clustering

《多粒度信息融合对比学习与多视图聚类》

简介信息

作者: Hengrong Ju , Yang Lu , Weiping Ding , Senior Member, IEEE, Wei Zhang , and Xibei Yang

文献类型: 期刊

影响因子: [8.9](#)

文献来源: TNNLS 2025

原文链接: <https://ieeexplore.ieee.org/document/11030281>

源码链接: <https://github.com/Luyangabc/MGCMVC>

关键词: [#multi-granularity](#) [#contrastive-multi-view-clustering](#)

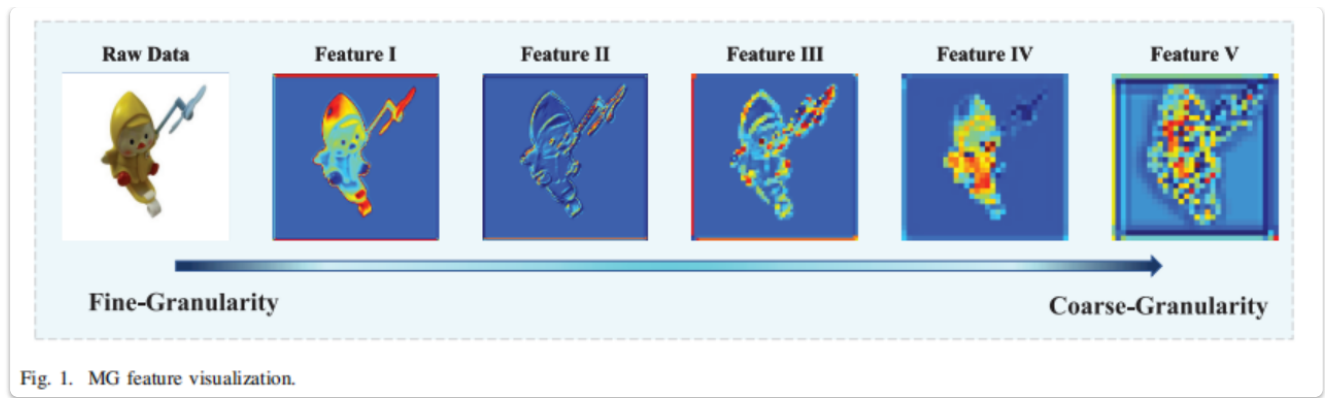


论文结构

- Abstract
- 1.Introduction
- 2.Related Work
- 3.Proposed Method
- 4.Experiments
- 5.Conclusion



研究问题



- 作者观察到：前神经网络架构，特征提取遵循从低级到高级的层次架构。
 - 例如，当学习图像特征时，神经网络最初会捕捉边缘、纹理等低级特征，然后提取物体形状、语义结构等高级特征。
- 然而，现有对比方法在实际应用时**仅采用高级信息进行聚类。低级信息的缺失限制了聚类的性能表现。**
- 同时，**量化不同视图的重要性**是对比多视图聚类仍然面临的关键挑战。
- 此外，在概率聚类空间中，**缺乏结构性信息**的特征描述进一步限制了当前方法的性能。

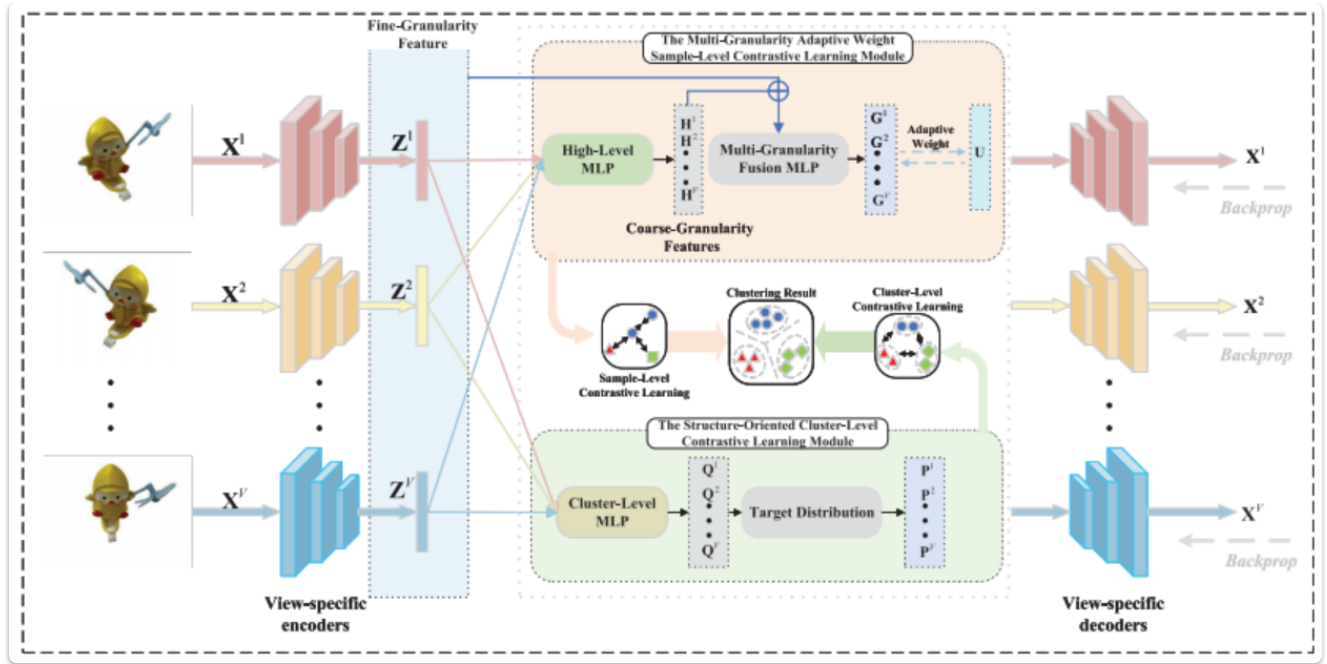


研究方法

为了解决以上问题，本文提出了MGCMVC（多粒度信息融合的对比多视图聚类），具体包含以下两个关键模块：

1. **多粒度自适应加权的样本对比学习机制**被提出用来**缓解因视图质量偏差**带来的性能下降。
2. **结构引导的聚类级对比学习**被设计来保存结构信息和跨视图一致性。

其实就是给MFLVC加了个权重。



具体实现

1. 保持现有自编码器架构不变，以学习每个视图的潜在表示。
2. 获取细粒度特征 Z^v 和粗粒度特征 H^v

1. 细粒度特征 Z^v 是从自编码器直接学习到的潜在表示
2. 粗粒度特征 H^v 是 Z^v 过一个共享的 MLP 学习到的新特征

$$G^v = F(Z^v; H^v; \phi^v) \\ = F([z_1^v, z_2^v, \dots, z_n^v]; [h_1^v, h_2^v, \dots, h_n^v]; \phi^v)$$

- 拼接粗细粒度特征，通过一层共享 MLP 学习到多粒度特征 G^v ，对其进行普通的对比学习（然而，这样没有考虑不同视图的质量，所以加了个权重）

$$l_{\text{sample}}^{v,m} = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{\text{sim}(g_i^v, g_i^m)/\tau_s}}{\sum_{i=1}^N \sum_{m=v,m} e^{\text{sim}(g_i^v, g_i^m)/\tau_s} - e^{1/\tau_s}}$$

- 本文融合了一个全局特征（拼接所有多粒度特征，过一层 MLP），计算每个视图的多粒度特征与全局特征的 Wasserstein 距离，因此上面的样本级对比学习可以重写为

$$GW^v = \frac{e^{-WD(G^v, U)}}{\sum_{v=1}^V e^{-WD(G^v, U)}}$$

$$L_S = \frac{1}{2} \sum_{v=1}^V \sum_{v \neq m} (GW^v + GW^m) \times l_{\text{sample}}^{v,m}$$

- 最后论文加了一个所谓软分配聚类级对比学习 (好像没啥区别啊)

$$p_{ij}^v = \frac{(q_{ij}^v)^2 / T_j}{\sum_{j'} (q_{ij'}^v)^2 / T_{j'}}$$

$$l_{\text{cluster}}^{v,m} = -\frac{1}{C} \sum_{j=1}^C \log \frac{e^{\text{sim}(p_j^v, p_j^m) / \tau_c}}{\sum_{c=1}^C \sum_{n=v,m} e^{\text{sim}(p_j^v, p_j^c) / \tau_c} - e^{1/\tau_c}}$$

$$L_C = \frac{1}{2} \sum_{v=1}^V \sum_{v \neq m} l_{\text{cluster}}^{v,m} + \sum_{v=1}^V \sum_{c=1}^C f_j^v \log f_j^v$$



实验结果

聚类结果

TABLE II
CLUSTERING PERFORMANCE OF ALL METHODS ON DATASETS COIL20, FASHION, DIGIT, CIFAR10, AND NOISYMNIST

Methods	COIL20			Fashion			Digit			Cifar10			NosiyMNIST		
	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR
DEMVC(2021)	0.850	0.965	0.850	0.848	0.871	0.848	0.940	0.949	0.940	0.412	0.468	0.412	0.781	0.826	0.781
CoMVC(2021)	0.870	0.960	0.870	0.867	0.872	0.881	0.985	0.974	0.985	0.993	0.980	0.993	0.903	0.935	0.903
MFLVC(2022)	0.990	0.992	0.990	0.992	0.98	0.992	0.995	0.987	0.995	0.993	0.979	0.993	0.994	0.981	0.994
CVCL(2023)	0.847	0.889	0.851	0.993	0.982	0.993	0.989	0.969	0.989	0.992	0.978	0.992	0.972	0.923	0.972
GCFAggMVC(2023)	1	1	1	0.989	0.9724	0.989	0.969	0.942	0.969	0.991	0.974	0.991	0.982	0.955	0.982
DealMVC(2023)	0.746	0.918	0.750	0.896	0.9542	0.896	0.865	0.927	0.890	0.182	0.147	0.182	0.995	0.983	0.995
MVCAN(2024)	1	1	1	0.834	0.847	0.838	0.988	0.972	0.988	0.994	0.984	0.994	0.981	0.959	0.981
Vital(2024)	1	1	1	0.971	0.941	0.971	0.998	0.995	0.998	0.909	0.804	0.910	0.984	0.954	0.984
CANDY(2024)	0.854	0.897	0.869	0.776	0.753	0.819	0.941	0.879	0.941	0.903	0.793	0.903	0.699	0.707	0.748
Ours	1	1	1	0.993	0.982	0.993	0.997	0.992	0.997	0.995	0.984	0.995	0.995	0.985	0.995

TABLE III
CLUSTERING PERFORMANCE OF ALL METHODS ON MNIST-USPS, BDGP, PROKARYOTIC, AND SYNTHETIC3D

Methods	MNIST-USPS			BDGP			Prokaryotic			Synthetic3d		
	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR
DEMVC(2021)	0.996	0.989	0.996	0.751	0.75	0.751	0.535	0.316	0.693	0.832	0.654	0.832
CoMVC(2021)	0.987	0.977	0.987	0.812	0.733	0.813	0.424	0.213	0.675	0.953	0.818	0.952
MFLVC(2022)	0.993	0.985	0.993	0.986	0.96	0.986	0.472	0.256	0.587	0.965	0.854	0.965
CVCL(2023)	0.997	0.991	0.997	0.992	0.973	0.992	0.514	0.223	0.568	0.967	0.87	0.967
GCFAggMVC(2023)	0.995	0.986	0.995	0.987	0.961	0.987	0.623	0.378	0.731	0.97	0.871	0.97
DealMVC(2023)	0.922	0.91	0.922	0.988	0.963	0.988	0.632	0.278	0.632	0.877	0.657	0.877
MVCAN(2024)	0.948	0.922	0.948	0.979	0.937	0.979	0.532	0.31	0.679	0.972	0.878	0.972
Vital(2024)	0.998	0.995	0.998	0.720	0.635	0.748	0.543	0.310	0.624	0.902	0.695	0.902
CANDY(2024)	0.941	0.879	0.941	0.874	0.758	0.874	0.575	0.273	0.626	0.785	0.413	0.785
Ours	0.997	0.991	0.997	0.993	0.977	0.993	0.675	0.391	0.764	0.982	0.914	0.982

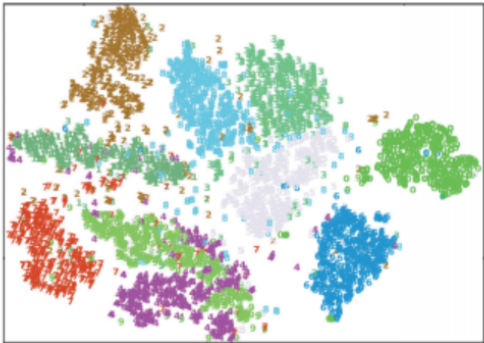
TABLE IV
CLUSTERING PERFORMANCE OF ALL METHODS ON CALTECH

Methods	Caltech-2V			Caltech-3V			Caltech-4V			Caltech-5V		
	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR
DEMVC(2021)	0.486	0.342	0.486	0.505	0.366	0.512	0.454	0.315	0.474	0.457	0.378	0.489
CoMVC(2021)	0.462	0.417	0.503	0.543	0.511	0.584	0.583	0.527	0.614	0.667	0.576	0.697
MFLVC(2022)	0.606	0.528	0.616	0.633	0.561	0.633	0.717	0.624	0.728	0.797	0.691	0.798
CVCL(2023)	0.616	0.53	0.628	0.723	0.632	0.736	0.781	0.699	0.789	0.843	0.737	0.843
GCFAggMVC(2023)	0.644	0.512	0.644	0.640	0.535	0.64	0.734	0.661	0.734	0.821	0.703	0.821
DealMVC(2023)	0.606	0.508	0.606	0.644	0.575	0.673	0.823	0.726	0.823	0.870	0.780	0.870
MVCAN(2024)	0.532	0.399	0.5743	0.862	0.767	0.86	0.896	0.807	0.896	0.860	0.769	0.860
Vital(2024)	0.649	0.506	0.649	0.675	0.579	0.675	0.664	0.57	0.664	0.672	0.49	0.672
CANDY(2024)	0.664	0.518	0.664	0.667	0.526	0.667	0.679	0.525	0.679	0.679	0.525	0.679
Ours	0.696	0.592	0.696	0.760	0.671	0.760	0.841	0.738	0.841	0.887	0.795	0.887

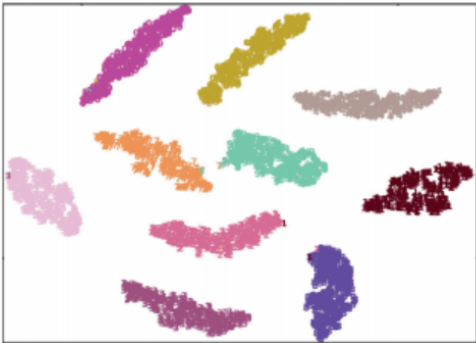
不同粒度结果对比

TABLE V
CLUSTERING PERFORMANCE OF DIFFERENT GRANULARITY FEATURES

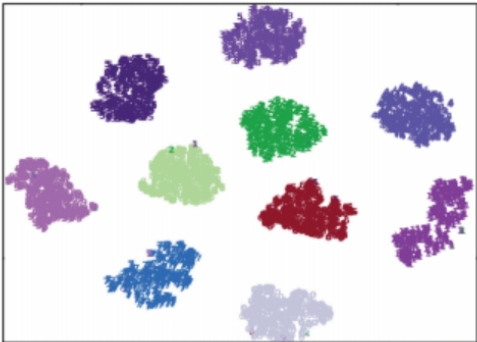
	Fashion			COIL20			Digt			MNIST-USPS			NoisyMNIST		
	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR
FG	0.827	0.860	0.850	0.922	0.967	0.944	0.980	0.956	0.980	0.994	0.984	0.994	0.954	0.912	0.954
CG	0.986	0.966	0.986	0.973	0.989	0.981	0.995	0.986	0.995	0.996	0.989	0.996	0.993	0.978	0.993
MG	0.993	0.982	0.993	1	1	1	0.997	0.992	0.997	0.997	0.991	0.997	0.995	0.984	0.995
	BDGP			Prokaryotic			Synthetic3d			Caltech-5V			Cifar10		
	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR
FG	0.924	0.862	0.924	0.548	0.276	0.681	0.849	0.625	0.849	0.867	0.773	0.867	0.707	0.695	0.711
CG	0.974	0.934	0.974	0.532	0.312	0.672	0.877	0.653	0.877	0.825	0.732	0.825	0.935	0.854	0.935
MG	0.994	0.981	0.994	0.672	0.411	0.770	0.982	0.914	0.982	0.887	0.795	0.887	0.995	0.984	0.995



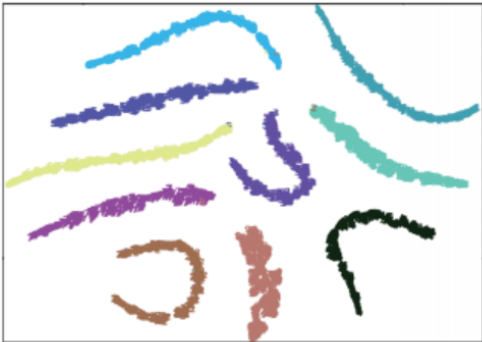
(a)



(b)



(c)



(d)

Fig. 3. Visualization of different granularity features on MNIST-USPS. (a) Space distribution of the raw data. (b) Space distribution of the FG feature Z . (c) Space distribution of the CG feature H . (d) Space distribution of the MG feature G .

消融实验

TABLE VI
ABLATION STUDY OF LOSS TERMS

	Loss Term			Prokaryotic			Caltech-5V			MNIST-USPS			Fashion			Cifar10			NoisyMnist		
	L_Z	L_S	L_C	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR
(a)		✓	✓	0.446	0.213	0.579	0.692	0.634	0.699	0.992	0.977	0.992	0.990	0.975	0.990	0.9929	0.98	0.9929	0.9938	0.9799	0.9938
(b)	✓		✓	0.459	0.174	0.570	0.867	0.786	0.867	0.993	0.981	0.993	0.991	0.978	0.991	0.9925	0.9792	0.9925	0.9951	0.9837	0.9951
(c)	✓	✓		0.588	0.314	0.682	0.525	0.512	0.385	0.413	0.592	0.419	0.744	0.743	0.744	0.71	0.7815	0.71	0.5896	0.7118	0.5957
	✓	✓	✓	0.672	0.411	0.77	0.887	0.795	0.887	0.997	0.991	0.997	0.993	0.982	0.993	0.9945	0.9842	0.9945	0.9954	0.9846	0.9954



创新不足

- 创新
 - 没啥创新 (Motivation有点意思)
- 局限
 -



感悟心得

- 思考

-



笔记摘抄

- As a fundamental approach to unsupervised data processing, clustering analysis holds substantial theoretical and practical value due to its ability to simplify complex data and transform disorder into structured patterns. 作为无监督数据处理的基础方法，聚类分析因其能够简化复杂数据，并将无序数据转换为结构化模式，在理论和实践层面都具有重要价值。
- Traditional clustering methods generally operate within a single view of the data, disregarding the intrinsic correlations among multiple views. 传统聚类方法通常旨在单一视角上处理数据，忽视了多视图之间的固有关联。
- [Xie et al.](#) 采用自编码器提取数据特征，并将其映射到潜在空间进行聚类，首次提出了神经网络损失和聚类损失相结合的深度聚类范式，为后续DMVC方法的发展提供了重要启发。
- Current contrastive methods primarily focuses on two levels of contrastive learning. 当前的对比方法主要分为两类
 - At the sample level, contrastive learning encourages similar samples to be drawn closer while pushing dissimilar samples further apart. 在样本级别，对比学习鼓励相似样本离得更近，同时将不相似的样本进一步推远离。
 - At the cluster level, it aims to enforce consistency in clustering distributions across multiview. 在聚类级别，对比学习强制多视图之间的聚类分布要保持一致。
- In the existing neural network architectures, **feature extraction follows a hierarchical process from low-level to high-level representations**. 当前神经网络架构，特征提取遵循从低级到高级的层次架构。
 - For instance, when learning image features, neural networks initially capture low-level attributes such as edges and textures, and as training progresses, they extract high-level features such as object shapes and semantic structures. 例如，当学习图像特征时，神经网络最初会捕捉边缘、纹理等低级特征，然后提取物体形状、语义结构等高级特征。